CAGLA AKIN

Data Science · Finance · Operations · Amazon · Carnegie Mellon University (PhD) · Imperial College London (Executive Education)

SUMMARY

I'm a data scientist working as a Finance Manager at Amazon. I turn business questions into models and pipelines that senior leadership acts on. I hold Tech Nation UK's Exceptional Talent endorsement in Digital Technology and do not require sponsorship.

EXPERIENCE

AMAZON.COM, Inc. | Seattle, US & London, UK — Aug 2019 - Present

Finance Manager, Supply Chain Optimization Technologies

- Built a global supply-chain simulation model in two weeks during COVID-19; its daily metrics became the single source of truth behind the SVP decision to cut delivery promises from 21 to 5 days, supporting 38% revenue growth in 2020.
- Developed Amazon's North America 1-Day delivery-speed forecasting model, backtested to 197 bps error against a ±500 bps threshold; it became leadership's basis for pursuing a $3.8B revenue opportunity.
- Architected the data infrastructure for Amazon's transportation cost anomaly-detection product, integrating 50+ tables into a 300GB library and co-designing its causal model with Core AI economists, uncovering $300M in free cash flow.
- Delivered a causal model quantifying the cost of labor-forecast error with the science team and automated its reporting into weekly business reviews, surfacing $78M in annual savings and realizing $11.6M as forecast error fell 4.2%.
- Designed a 10% inventory-holdout A/B test and turned its results into the business metrics for capacity control systems, quantifying $40M in lost Q4 revenue and driving changes to the inventory-health rule engines.
- As acting Senior Product Manager, owned the standardization and automation of key metadata feeding Amazon's financial and operational planning platforms, including a global facility dataset that cut manual workloads 60% and saved $0.5M a year.
- Owned financial strategy for $10B of EU and Japan inventory; audited the cost inputs of the forecast run alongside the science team's Bayesian VAR model, identifying $50M in clearance opportunity and actioning $10M.

Previous titles: Senior Financial Analyst (London, UK) · Financial Analyst (Seattle, US)

CODE YOUR FUTURE | London, UK — Jan 2025 - Present

Data Scientist (Volunteer, part-time)

- Run the charity's performance analytics for the CEO and COO, building cohort-funnel and volume/rate decomposition analyses that explain year-over-year change and separate structural drivers from one-off effects.
- As Product Lead, ran Scrum for a team of 9 software and QA engineers building the CYF Dashboard, the platform behind the charity's CRM and learning management, serving 300+ active users per batch and thousands of applicants.

CARNEGIE MELLON UNIVERSITY | Pittsburgh, PA, US — Aug 2016 – May 2019

PhD Candidate/Doctoral Researcher

- Applied difference-in-differences and quasi-natural experiment designs across three empirical papers on regulatory interventions in banking, municipal bond and energy markets, earning the American Accounting Association's Best Doctoral Student Paper Award.

DELOITTE TOUCHE TOHMATSU | Istanbul, Turkey — Jun 2015 – Jul 2016

Financial Advisory Associate

- Advised C-suite executives on M&A decisions, turning due diligence and valuation analysis into deal recommendations.

EDUCATION

IMPERIAL COLLEGE BUSINESS SCHOOL | London, UK — 2025 – 2026

Executive Education in Data Science

CARNEGIE MELLON UNIVERSITY | Pittsburgh, PA, US — 2016 – 2019

Tepper School of Business, MSc. in Industrial Administration (Accounting)

(Completed PhD coursework and research before earning a Master's degree)

BILKENT UNIVERSITY | Ankara, TR — 2011 – 2015

Bachelor of Science in Business Administration, Minor in Economics

UNIVERSITY OF BOCCONI | Milan, IT — 2014

Exchange Student, Finance and Economics

SKILLS

Causal Inference & Experimentation: difference-in-differences, regression discontinuity, A/B test design, fixed-effects models
Machine Learning & Forecasting: XGBoost, random forests, regression, clustering, simulation, backtesting; NLP (sentiment, topic modelling), LLM prompt engineering
Data Engineering & Analytics: SQL ETL pipelines, data integration across sources, metrics design, Tableau, QuickSight
Product & Leadership: roadmap ownership, Agile/Scrum, executive communication, cross-functional partnership
Tools: Python (pandas, NumPy, scikit-learn, statsmodels), SQL, AWS SageMaker, Git, MS Excel, Claude Code

DATA SCIENCE PROJECTS & TRAINING

Applied AI Bootcamp | AllWomen Tech

- Hands-on training in machine learning, deep learning, generative AI, and NLP using Python.
- Worked with pretrained LLMs (Llama 2, Hugging Face) for NLP tasks including text classification and summarisation; applied prompt engineering to adapt models for business use cases.
- Studied AI governance and responsible AI: Bias and fairness metrics, protected attributes, model explainability (SHAP, global vs local), and the regulatory landscape including the EU AI Act, GDPR compliance, model cards and dataset documentation.

Spotify Hit Predictor | Machine Learning Capstone

- XGBoost model predicting song chart success from ~170K tracks; iterated linear regression → random forest → XGBoost (~80% accuracy), finding artist reputation matters far more than audio characteristics.
- Deployed to an AWS SageMaker inference endpoint and published as an interactive project site (link).

Suno AI Brand Monitor | NLP Project

- NLP brand-monitoring prototype analysing public conversations with VADER sentiment and keyword-guided topic modelling.
- Found ~68% positive sentiment, with copyright and legal exposure as the key reputational risk.

AWARDS & RECOGNITION

- Amazon Future Leader (AFFL) Constellation Award | Amazon — Jul 2025
- Exceptional Talent in Digital Technology | endorsed by Tech Nation UK — Apr 2025
- Best Doctoral Student Paper Award | American Accounting Association — May 2019
- William Larimer Fellowship | Carnegie Mellon University (5-year doctoral fellowship) — Aug 2016

PUBLICATIONS

- Akin, C. (2025). Causal Inference in E-Commerce and Technology Business. Harvard Business Review Türkiye (link).
- Akin, C. (2019). Consolidation of Securitization Entities and the Effects on Charitable Lending. Journal of Management Policy and Practice, 20(4) (link).
- Akin, C. Financial Disclosure in Municipal Bond Markets. Working Paper.
- Research contribution (acknowledged): Intended and Unintended Consequences of Facility-Level Mandatory CSR Disclosure. Journal of Accounting, Auditing & Finance (2026) (link).

PUBLIC SPEAKING & TEACHING

- Panelist & Moderator, NextGen Finance Transformation Conference | London — Jun 2026
- Judge, McGinnis Venture Competition | Carnegie Mellon University — Jan 2025, 2026
- Guest Speaker, career events | Bilkent University and Koç University, Türkiye — Mar 2025
- Guest Lecturer, Causal Inference Methods | Aalto School of Business / Helsinki GSE — Nov 2024
- Moderator & Presenter, Capital Markets Research Session | Mid-Atlantic Regional Meeting — May 2019
- Presenter, Annual Accounting Mini Conference | Carnegie Mellon University — 2016, 2017, 2018
- Teaching Assistant for MBA and MSc in Information Systems | Carnegie Mellon University — 2016, 2017, 2018

LEADERSHIP & COMMUNITY

- **Tech Startup Mentor, ITU Çekirdek:** Advise seed-stage tech start-ups on digital strategy and operational scaling.
- **Business Mentor, Young Enterprise UK:** Coach high-school enterprise teams through the programme year and judge the national final, assessing the top companies among 50+ competing teams.
- **DJ & Producer, Sunrise Fiancée:** Curate and perform melodic house and techno across London radio, club, and hospitality settings, including a residency at Hart Shoreditch Hotel; programme live sets to venue, audience, and moment.